

1 Mathematics

2 Notes

Mathematics (1)

1.1 Trigonometry

$$\sin(v+w) = \sin v \cos w + \cos v \sin w$$
$$\cos(v+w) = \cos v \cos w - \sin v \sin w$$
$$\tan(v+w) = \frac{\tan v + \tan w}{1 - \tan v \tan w}$$
$$\sin v + \sin w = 2 \sin \frac{v+w}{2} \cos \frac{v-w}{2}$$
$$\cos v + \cos w = 2 \cos \frac{v+w}{2} \cos \frac{v-w}{2}$$

$$(V+W) \tan(\frac{v-w}{2}) = (V-W) \tan(\frac{v+w}{2})$$

V, W are sides opposite to angles v, w . $a \cos x + b \sin x = r \cos(x - \phi)$
 $a \sin x + b \cos x = r \sin(x + \phi)$
where $r = \sqrt{a^2 + b^2}, \phi = \text{atan2}(b, a)$.

1.2 Geometry

1.2.1 Triangles

Side lengths: a, b, c

Semiperimeter: $p = \frac{a+b+c}{2}$

Area: $A = \sqrt{p(p-a)(p-b)(p-c)}$

Circumradius: $R = \frac{abc}{4A}$

Inradius: $r = \frac{A}{p}$

Length of median (divides triangle into two equal-area triangles): $m_a = \frac{1}{2}\sqrt{2b^2 + 2c^2 - a^2}$

Length of bisector (divides angles in two): $s_a = \sqrt{bc[1 - (a/(b+c))^2]}$

Law of sines, cosines & tangents: $\frac{\sin \alpha}{a} = \frac{\sin \beta}{b} = \frac{\sin \gamma}{c} = \frac{1}{2R}.....(1)$

$a^2 = b^2 + c^2 - 2bc \cos \alpha.....(2)$

$\frac{a+b}{a-b} = \frac{\tan((\alpha+\beta)/2)}{\tan((\alpha-\beta)/2)}.....(3)$

1.2.2 Quadrilaterals

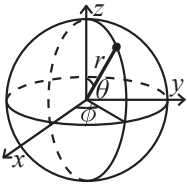
With side lengths a, b, c, d , diagonals e, f , diagonals angle θ , area A and magic flux $F = b^2 + d^2 - a^2 - c^2$:

$$4A = 2ef \cdot \sin \theta = F \tan \theta = \sqrt{4e^2 f^2 - F^2}$$

For cyclic quadrilaterals the sum of opposite angles is 180° , $ef = ac + bd$, and $A = \sqrt{(p-a)(p-b)(p-c)(p-d)}$

1 1.2.3 Spherical coordinates

2



$$x = r \sin \theta \cos \phi \quad r = \sqrt{x^2 + y^2 + z^2}$$
$$y = r \sin \theta \sin \phi \quad \theta = \text{acos}(z/\sqrt{x^2 + y^2 + z^2})$$
$$z = r \cos \theta \quad \phi = \text{atan2}(y, x)$$

1.3 Sums

$$c^a + c^{a+1} + \dots + c^b = \frac{c^{b+1} - c^a}{c - 1}, \quad c \neq 1$$

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(2n+1)(n+1)}{6}$$

$$1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4}$$

$$1^4 + 2^4 + 3^4 + \dots + n^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}$$

$$1^4 + 2^4 + 3^4 + \dots + n^4 = S_2 \times \frac{3n^2 + 3n - 1}{5} = S_4$$

$$b \sum_{k=0}^{n-1} (a+kd)r^k = \frac{ab - (a+nd)br^n}{1-r} + \frac{dbr(1-r^n)}{(1-r)^2}$$

To compute $1^k + \dots + n^k$ in $\mathcal{O}(k \lg k + k \lg MOD)$ compute first $t = k + 2$ sums $y_1, \dots, y_t$, then interpolate. Let $P = \prod_{i=1}^t (n-i)$. Then answer for n is

$$\sum_{i=1}^t \frac{P}{n-i} \cdot \frac{(-1)^{t-i} y_i}{(i-1)!(t-i)!}$$

Also $S_k = \frac{1}{k+1} \sum_{j=0}^k (-1)^j \binom{k+1}{j} B_j n^{k+1-j}$ where B_i are Bernoulli numbers.

1.4 Bitwise Formulas

Some properties of bitwise operations:

$$a|b = a \oplus b + a\&b$$
$$a \oplus (a\&b) = (a|b) \oplus b$$
$$b \oplus (a\&b) = (a|b) \oplus a$$
$$(a\&b) \oplus (a|b) = a \oplus b$$

Addition:

$$a + b = a|b + a\&b$$
$$a + b = a \oplus b + 2(a\&b)$$

Subtraction:

$$a-b=((a\&b)\oplus((a|b)\oplus a))$$
$$a-b=((a|b)\oplus b)-((a|b)\oplus a)$$
$$a-b=(a\oplus(a\&b))-((a|b)\oplus a)$$
$$a-b=(a\oplus(a\&b))-(b\oplus(a\&b))$$
$$a-b=((a|b)\oplus b)-(b\oplus(a\&b))$$

Working rules for logarithms:

1.

$\log_b(mn)=\log_b m+\log_b n$
2.

$\log_b\left(\frac{m}{n}\right)=\log_b m-\log_b n$
3.

$\log_b(m^r)=r\log_b m$
4.

$\log_b 1=0$
5.

$\log_b b=1$
6.

$\log_b m=\frac{\log_a m}{\log_a b}$

1.5 Trivia

Pythagorean triples: The Pythagorean triples are uniquely generated by $a=k\cdot(m^2-n^2)$, $b=k\cdot(2mn)$, $c=k\cdot(m^2+n^2)$ with $m>n>0$, $k>0$, $\gcd(m,n)=1$, both m,n not odd.
Primes: $p=962592769$ is such that $2^{21}\mid p-1$, which may be useful. For hashing use 970592641 (31-bit number), 31443539979727 (45-bit), 3006703054056749 (52-bit). There are 78498 primes less than 1 000 000.

Primitive roots modulo n exists iff $n=1,2,4$ or, $n=p^k,2p^k$ where p is an odd prime. Furthermore, the number of roots are $\phi(\phi(n))$.

To Find Generator g of M , factor $M-1$ and get the distinct primes p_i . If $g^{(M-1)/p_i}\neq 1(MODM)$ for each p_i then g is a valid root. Try all g until a hit is found (usually found very quick).

Esitmates: $\sum_{d|n} d=O(n\log\log n)$.

Prime count: 5133 upto 5e4. 9592 upto 1e5. 17984 upto 2e5. 78498 upto 1e6. 5761455 upto 1e8.

max NOD $\leq n$: 100 for $n=5e4$. 500 for $n=1e7$. 2000 for $n=1e10$. 200 000 for $n=1e19$.

max Unique Prime Factors: 6 upto 5e5. 7 upto 9e6. 8 upto 2e8. 9 upto 6e9. 11 upto 7e12. 15 upto 3e19.

Quadratic Residue: $(\frac{a}{p})$ is 0 if $p|a$, 1 if a is a quadratic residue, -1 otherwise. Euler: $(\frac{a}{p})=a^{(p-1)/2}(\bmod p)$ (prime). Jacobi: if $n=p_1^{e_1}\cdots p_k^{e_k}$ then $(\frac{a}{n})=\prod(\frac{a}{p_i})^{e_i}$.

Chicken McNugget. If a,b coprime, there are $\frac{1}{2}(a-1)(b-1)$ numbers not of form $ax+by$ ($x,y\geq 0$), the largest being $ab-a-b$.

Notes (2)

2.1 Fibonacci

Let A,B , and n be integers. Define $k=A-B$.

$$F_A F_B=F_{k+1} F_A F_A+F_k F_A F_{A-1}$$

$$\sum_{i=0}^n F_i^2=F_{n+1} F_n$$

$$\sum_{i=0}^n F_i F_{i+1}=F_{n+1}^2-(-1)^n$$

$$\sum_{i=0}^n F_i F_{i-1}=\sum_{i=0}^{n-1} F_i F_{i+1}$$

$$\gcd(F_m,F_n)=F_{\gcd(m,n)}$$

$$\sum_{0\leq k\leq n} \binom{n-k}{k}=Fib_{n+1}$$

$$\gcd(F_n,F_{n+1})=\gcd(F_n,F_{n+2})=\gcd(F_{n+1},F_{n+2})=1$$

2.2 More Number Theory

$$\sum_{k=1}^n \frac{1}{\gcd(k,n)}=\sum_{d|n} \frac{1}{d} \phi\left(\frac{n}{d}\right)=\frac{1}{n} \sum_{d|n} d \phi(d)$$

$$\sum_{k=1}^n \frac{k}{\gcd(k,n)}=\frac{n}{2}\cdot \frac{1}{n} \sum_{d|n} d \phi(d)$$

$$\sum_{k=1}^n \frac{n}{\gcd(k,n)}=2 \sum_{k=1}^n \frac{k}{\gcd(k,n)}-1,\quad n>1$$

$$\sum_{i=1}^n \sum_{j=1}^n [\gcd(i,j)=1]=\sum_{d=1}^n \mu(d) \left\lfloor \frac{n}{d} \right\rfloor^2$$

$$\sum_{i=1}^n \sum_{j=1}^n \gcd(i,j)=\sum_{d=1}^n \phi(d) \left\lfloor \frac{n}{d} \right\rfloor^2$$

$$\sum_{i=1}^n \sum_{j=1}^n i\cdot j\,[\gcd(i,j)=1]=\sum_{i=1}^n \phi(i)\,i^2$$

$$\sum_{i=1}^n \sum_{j=1}^n \text{lcm}(i,j)=\sum_{l=1}^n \left(\frac{(1+\lfloor n/l\rfloor)\lfloor n/l\rfloor}{2}\right)^2$$

2.2.1 Permutations

Factorial Values

$\frac{n}{n!}$	1	2	3	4	5	6	7	8	9	10
	1	2	6	24	120	720	5040	40320	362880	3628800

$\frac{n}{n!}$	11	12	13	14	15	16	17
	4.0e7	4.8e8	6.2e9	8.7e10	1.3e12	2.1e13	3.6e14

$\frac{n}{n!}$	20	25	30	40	50	100	150	171
	2e18	2e25	3e32	8e47	3e64	9e157	6e262	>DBL.MAX